

React

```
import React, { useState } from "react";
import { Button } from "react-bootstrap";
// import styles from "../styles/modules.css"

export const container = {
  backgroundColor: "blue",
  padding: "10px",
  color: "red",
};

const ControlledComponent = () => {
  const [name, setName] = useState("");
  const [age, setAge] = useState(0);

  const handleSubmit = (e) => {
    e.preventDefault();
    console.log(name);
    console.log(age);
    setName("");
    setAge(0);
  };

  return (
    <div>
      <form action="">
        Name :{" "}
        <input
          type="text"
          value={name}
          onChange={(e) => setName(e.target.value)}
          placeholder="Enter your name"
        />
        <br />
        Age :{" "}
        <input
          type="number"
          value={age}
          onChange={(e) => setAge(e.target.value)}
          placeholder="Enter your age"
        />
        <br />
        { /* <button type="Submit" onClick={(e) => handleSubmit(e)}>
          Submit
        </button> */ }
      </div>
    );
  };
};
```

```

        <Button variant="primary">Bootstrap Button</Button>
      </form>
    </div>
  );
};

export default ControlledComponent;

```

```

import React, { useEffect } from 'react'
import { container } from './ControlledComponent';

const useDocumentTitle = (title) => {
  useEffect(() => {
    document.title = title;
  }, [title]);
};

const CustomHook = () => {

  useDocumentTitle("Redmi");

  return (
    <div style={container}>
      <h1>Custom hook</h1>

    </div>
  )
}

export default CustomHook

```

```

import axios from "axios";
import React, { useEffect, useState } from "react";
import { api } from "./api";

const AxiosComponent = () => {
  const [data, setData] = useState([]);
  const [name, setName] = useState("");

  const fetchData = async () => {
    const result = await api.get("/users");

```

```

    console.log(result.data);
    setData(result.data);
  };

  useEffect(() => {
    fetchData();
  }, []);

  const handleSubmit = async (e) => {
    e.preventDefault();

    try {
      const result = await axios.post(
        "https://jsonplaceholder.typicode.com/users",
        { name },
      );
      console.log(result.data);
      setData((prev) => [...prev, result.data]);
      setName("");
    } catch (error) {
      console.log(error);
    }
  };

  const Updateuser = async () => {
    try {
      const result = await axios.put(
        "https://jsonplaceholder.typicode.com/users/1",
        { name: "REDMI" },
      );

      console.log(result);

    } catch (error) {
      console.log(error);
    }
  };

  return (
    <div>
      <h2>Axios</h2>
      <button onClick={Updateuser}>Update</button>

      { /* list rendering */ }
      <form onSubmit={handleSubmit}>
        <input
          type="text"
          value={name}

```

```

        onChange={(e) => setName(e.target.value)}
        placeholder="Enter your name"
      />

      <button type="submit">Add</button>
    </form>

    {data.map((ele) => {
      return (
        <div key={ele.id}>
          <p>{ele.name}</p>
        </div>
      );
    })}
  </div>
);
};

export default AxiosComponent;

```

```

import axios from "axios";

export const api = axios.create({
  baseURL: "https://jsonplaceholder.typicode.com",
});

```

Theory

1. Controlled Components

- **Definition:** Form inputs whose values are controlled by React state.
- **Key points:**
 - `value={state}` binds input to state.
 - `onChange={(e) => setState(e.target.value)}` updates state.
 - Ensures React has full control over form data.

2. Custom Hooks

- **Definition:** Reusable functions that encapsulate logic using hooks.
- **Example:** `useDocumentTitle(title)` updates the document title.

- **Benefits:**
 - Code reusability.
 - Cleaner components.
 - Separation of concerns.

3. Axios Integration

- **Definition:** Promise-based HTTP client for API calls.
- **Common methods:**
 - `axios.get(url)` → fetch data.
 - `axios.post(url, body)` → create data.
 - `axios.put(url, body)` → update data.
 - `axios.delete(url)` → delete data.
- **Best practice:** Create a reusable api instance with `axios.create()`.



Coding Round Questions

- Controlled components → What are controlled components and why are they useful?
- Custom hooks → How do you create and use custom hooks?
- Axios → How do you perform CRUD operations with Axios?
- API integration → How do you integrate external APIs in React?
- Form handling → How do you handle forms in React using state?
- Error handling → How do you handle API errors with Axios?



Machine Coding Round Challenge

Problem: User Management App

Requirements:

1. Create a form with controlled inputs (name, email).
2. Use Axios to fetch users from an API.
3. Add new users via POST.
4. Update existing users via PUT.
5. Delete users via DELETE.
6. Display users in a list with edit/delete buttons.